

Jimmy Van Hout

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Arlington, VA

Education

University of Maryland, College Park, MD, *May 2022*

- **Bachelor of Science in Computer Science**
- Minor in Spanish Language, Culture, and Professional Contexts

Experience

Embedded Software Engineer

EmbedTek (Subcontractor), *February 2025 - Present*

- Work directly with President & CTO of a medical device manufacturer to design software and hardware for use in hospitals and nursing homes
- Process data from multiple sensors using multithreading on both cores of an embedded ESP32 microcontroller to achieve high throughput
- Develop software to control mechanical devices critical for patient health and safety
- Assist in selecting and designing hardware, including PCB, to enable software control and proper functioning of device
- Develop software on embedded computer or remote server to store, visualize, and process data

Full-Stack Software Engineer

Quadrofoil (Contractor), *August 2025 - September 2025*

- Worked with client to identify and solve an issue, gather requirements, obtain product feedback from stakeholders, and deliver a product according to clients' requirements
- Designed website according to client's specifications

Full-Stack Software Engineer

ScheduleAI, *June 2023 - January 2025*

- Developed a SaaS web application and artificial intelligence algorithm for automated employee scheduling
- Developed backend using Python, Flask, PostgreSQL, Amazon Cognito API; frontend using JavaScript, HTML, CSS

- Implemented server-side user authentication and authorization, including the use of OAuth 2.0, Amazon Cognito, JWTs, and cryptographic libraries
- Develop HTTP REST APIs for programmatic access of data

Software Engineer

Orangetheory Fitness (Consultant), *August 2022 - November 2022*

- Developed software to analyze and visualize customer data and resource utilization to optimize resource allocation and reduce expenses

Technical Skills

- **Programming Languages:** Python, JavaScript, C, C++, Java, C#
- **Web:** Python, Flask, JavaScript, Node.JS, HTML, CSS, React, Vue, Gunicorn, NGINX, Linux, Figma
- **Embedded:** ESP32, STM32, FreeRTOS, esp-idf, Arduino, Raspberry Pi, Linux
- **Cloud Computing/SaaS:** AWS (EC2, Cognito, Lightsail, Route53, SES, etc.), GCP, Cloudflare, Azure, Stripe
- **Security:** Authentication and authorization, OAuth 2.0, JWTs, cryptographic libraries, NIST Secure Software Development Framework (SSDF)
- **Parallel and Multithreaded Programming:** HPC cluster at UMD, MPI, OpenMP, CUDA, Hatchet, Python multiprocessing module, Java Thread library
- **Databases:** PostgreSQL, SQLite
- **Version Control:** Git, GitHub
- **Operating Systems and Shells:** Linux (Debian, Ubuntu, Raspberry Pi OS), FreeRTOS, Mac OS, Windows, Bash

Selected Technical Projects

Temperature and Pressure Sensor Program

C, Python

Read temperature and pressure from a MS5839-02BA sensor via I2C with an ESP32 microcontroller and output the data over serial.

GitHub: https://github.com/JimmyVanHout/temperature_pressure_sensor